Online Resource 1: Additional Reproducibility Details

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Article title: Budgeted Page-Level Preprocessing Selection for Vietnamese Administrative OCR: A Controlled Comparison of Learned, Retrieval-Based, and Static Shortlists
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Caption: Additional dataset summary, sampled-pool oracle improvement, offline OCR evaluation loop, preprocessing search-space table, surrogate feature groups, nested-CV surrogate model quality, nested GroupKFold protocol details and stepwise algorithm, policy-agnostic headroom controls, Vietnamese best-of-K curves, strict policy-matched and rank-only oracle-free selector results, SROIE intermediate-budget results omitted from the main manuscript, post hoc standalone-feature sensitivity, and contour-suppression schematic referenced from the main manuscript.

1 Dataset Summary

Table 1: Dataset summary.

Aspect	Vietnamese administrative documents	SROIE receipt images
Dataset/domain	Vietnamese government administrative documents	SROIE receipt images
Source	Public Vietnamese Government document portal: `https://chinhphu.vn/he-thong-van-ban`	ICDAR 2019 SROIE benchmark
Images	300 first-page images	612 images after filename-level, exact-content and same-receipt filtering
Time span	2015–2024, balanced by year	Not applicable
Reference text	Manually transcribed from images without consulting OCR; subsequent targeted checks; frozen v3	Benchmark-provided full-text transcriptions; casefolded for primary CER
OCR recognizers	VietOCR, Tesseract, EasyOCR	Tesseract, EasyOCR

The final Vietnamese cohort preserves 30 first pages per year. An exact-content audit removed two duplicated selections and replaced them within their corresponding 2018 and 2024 strata before all reported Vietnamese experiments were recomputed; the public cohort manifest records final identifiers and image/reference SHA-256 hashes.

SROIE filtering first removed both members of five byte-identical pairs (626 to 616 receipts). A later screen of all 189,420 remaining image pairs produced 2,027 candidates. Targeted image review resolved two same-receipt pairs and 2,025 different-receipt comparisons: 49 direct pair reviews and 1,978 comparisons using image-verified transaction identifiers. This is not exhaustive duplicate-recall certification or a second independent human audit. Both members of the two confirmed pairs were excluded, leaving 612 pages and 30,600 candidate rows. All affected SROIE surrogate and nested selector/control training layers were rerun. New GroupKFold sizes are 123/123/122/122/122; 593 retained pages change numerical fold label relative to the 616-page cohort. Both metrics and Full/Drop share the new folds. Consequently, old/new differences include changed partitions, not an isolated duplicate-removal effect. Original outputs and exclusion/hash records remain available separately.

2 Sampled-Pool Oracle Improvement

Table 2: Sampled-pool oracle improvement over exact NoPre. Relative improvement is computed per page as (NoPre − SampOracle)/NoPre.

Setting	Recognizer	Pages improved	Mean rel.	Median rel.
Raw	VietOCR	288/300 (96.0%)	26.2%	23.7%
Raw	Tesseract	293/300 (97.7%)	36.4%	37.4%
Raw	EasyOCR	271/300 (90.3%)	15.4%	14.2%
Contour	VietOCR	282/300 (94.0%)	28.6%	28.0%
Contour	Tesseract	287/300 (95.7%)	39.4%	41.4%
Contour	EasyOCR	270/300 (90.0%)	16.7%	15.7%
SROIE	Tesseract	597/612 (97.5%)	32.4%	32.1%
SROIE	EasyOCR	585/612 (95.6%)	15.9%	13.8%

3 Offline OCR Evaluation Loop

Input: Page images and references $\mathcal{D}$; candidate pool Θ; recognizer set E; optional contour transform C.
Output: Collected data table with one row for each page i and preprocessing setting θ_j, containing page features, preprocessing settings, and observed CER values as regression targets.

1. For each page x_i, choose the page representation $\tilde{x}_i$: x_i for raw experiments or $C(x_i)$ for contour-suppressed experiments.
2. Extract OCR-free page features $f(\tilde{x}_i)$ once for that page representation.
3. For each candidate $\theta_j \in \Theta$, apply P_{θ_j} to $\tilde{x}_i$.
4. For each recognizer $e \in E$, run the initial shared-detection and backend-specific crop-processing pipeline and compute $y_{i,j,e}$ by page-level CER against r_i.
5. Store $z_{i,j}$, θ_j, page/fold identifiers, and CER targets $y_{i,j,e}$.

4 Preprocessing Search Space

Table 3: Page-level preprocessing search space.

Parameter	Range	Off value	Role
`gaussian_ksize`	{0, 3, 5, 7}	0	denoising
`bilateral_d`	{0, . . . , 9}	0	edge-preserving denoising
`clahe_clip`	[0, 4.0]	0	local contrast enhancement
`gamma`	[0.5, 1.5]	1.0	brightness correction
`sharpen`	[0, 2.0]	0	sharpening strength

A post hoc check with OpenCV 4.10 and the fixed bilateral-filter sigmas found that encoded positive diameters 1–3, 4–5, 6–7, and 8–9 gave identical outputs within each group; `bilateral_d=0` is off. The original 50-candidate pool is retained despite this observed parameter redundancy.

A post hoc projection-profile diagnostic on all 300 raw Vietnamese pages gave median and maximum absolute skew estimates of 0.15° and 1.40°. It searched ±5° in 0.05° steps on padded Otsu ink masks (width capped at 1,000 pixels), maximizing squared adjacent differences in horizontal projection. These estimator-dependent values do not establish deskew's effect on OCR, which was not tested. The stored Hough feature and fitted models were unchanged. Script, configuration, per-page estimates, score curves and hashes accompany the public release.

5 Surrogate Feature Groups

Table 4: Surrogate feature groups.

Feature group	Features	Purpose
Base image features	blur, contrast, brightness, `noise`, edge density, entropy, dynamic range	blur is Laplacian variance; `noise` is Laplacian standard deviation, its square root, not an independent noise estimate
Document-aware features	text density, skew angle, connected-component count, aspect ratio, estimated line count	capture page structure not represented by intensity statistics alone
Preprocessing parameters	`gaussian_ksize`, `bilateral_d`, `clahe_clip`, `gamma`, `sharpen`	describe the candidate preprocessing setting being scored
Interaction terms	blur × `gaussian_ksize`, contrast × `clahe_clip`, brightness × `gamma`, noise × `bilateral_d`, entropy × `sharpen`	encode predefined relationships between page properties and preprocessing controls

6 Nested-CV Surrogate Model Quality

Table 5: Nested-CV surrogate model quality. Spearman values are reported as mean ± standard deviation across outer folds. The final refit model is selected and fitted on all available pages; reported metrics are out-of-fold nested-CV results and do not use this final model.

Setting	Recognizer	Final refit model	MAE	R^2	Global Spearman	Per-image Spearman
Raw	VietOCR	XGBoost	0.021	0.471	0.593 ± 0.080	0.622 ± 0.018
Raw	Tesseract	XGBoost	0.046	0.589	0.678 ± 0.081	0.510 ± 0.013
Raw	EasyOCR	LightGBM	0.039	0.482	0.666 ± 0.066	0.766 ± 0.016
Contour	VietOCR	Random Forest	0.020	0.467	0.576 ± 0.064	0.623 ± 0.025
Contour	Tesseract	LightGBM	0.045	0.549	0.612 ± 0.078	0.551 ± 0.013
Contour	EasyOCR	XGBoost	0.037	0.523	0.668 ± 0.045	0.781 ± 0.033
SROIE	Tesseract	XGBoost	0.092	0.547	0.697 ± 0.020	0.620 ± 0.018
SROIE	EasyOCR	XGBoost	0.066	0.716	0.677 ± 0.048	0.661 ± 0.015

7 Nested GroupKFold Protocol

8 Nested GroupKFold Evaluation Algorithm

Input: Collected data table; candidate pool Θ; recognizer e; K values $\{3, 5, 10, 15\}$; model-family set $\mathcal{H}$.
Output: Out-of-fold page-level Surrogate K, Random K, FixedTopK, NN-K, FixedBest, NoPre, and sampled-oracle results.

1. Split pages into five outer GroupKFold folds, keeping all 50 candidate rows from the same page in the same fold.
2. For each outer fold, select $h_e \in \mathcal{H}$ by inner GroupKFold validation MAE on the outer-training pages.
3. Refit the selected model on all outer-training rows and score all $\theta_j \in \Theta$ for each held-out page.
4. Compute Surrogate K, Random K, FixedTopK, NN-K, FixedBest, NoPre, and sampled-oracle references using the definitions in the main manuscript.

Table 6: Nested GroupKFold evaluation protocol.

Component	Setting	Purpose
Outer evaluation split	5-fold GroupKFold grouped by page image	prevents rows from the same page appearing in both training and test data
Outer test unit	held-out pages, with all 50 candidates kept together	produces out-of-fold page-level evaluation
Inner model selection	5-fold GroupKFold on each outer training split	selects the surrogate model family by validation MAE without using outer test pages
Reported aggregation	held-out page-level results combined across the five outer folds	provides one out-of-fold result per page for CER summaries and statistical tests
Final refit	model family selected and refit on all available pages	final model is not used to compute reported cross-validation metrics

5. Pool held-out page-level results across the five outer folds.
6. Compute mean CER, bootstrap confidence intervals, and two-sided Wilcoxon signed-rank tests with SciPy `zero_method=wilcox` (zero paired differences excluded from signed ranks).
7. Apply Holm correction within each policy-contrast/evaluation-block family; retain both raw and adjusted p-values.

The inner loop selects the surrogate model family by candidate-level validation MAE. This preserves test-fold separation but is only indirectly aligned with the downstream within-page ranking and best-of-K objective.

9 Policy-Agnostic Headroom Controls

For each held-out page and budget, the main manuscript defines FixedTopK-to-oracle headroom as $H = F - O$. The coupled outcomes $F - P$, for policy P, share F with H and satisfy $F - P \leq F - O$. We therefore recomputed the correlation for Surrogate, Random, and NN-K and added direct policy-to-policy controls that do not contain FixedTopK. At every budget, all eight dataset/setting/recognizer configurations have significantly positive coupled correlations for all three policies after Holm correction within each policy's 32-correlation family. Table 7 gives the per-budget breakdown of the additional controls.

Table 7: Per-budget headroom-control breakdown. Random $r >$ Surrogate r compares the two coupled correlations. The remaining columns count significantly positive Spearman correlations after Holm correction within each direct outcome's 32-correlation family. Each budget contains eight dataset/setting/recognizer configurations.

K	Coupled positive, each policy	Random $r >$ Surrogate r	Random − Surrogate	Random − NN-K	NN-K − Surrogate
3	8/8	7/8	0/8	0/8	0/8
5	8/8	7/8	0/8	0/8	0/8
10	8/8	8/8	0/8	0/8	0/8
15	8/8	8/8	0/8	0/8	0/8

Across budgets, coupled correlations remain positive and significant in 32/32 configurations for each policy; Random has a larger correlation than Surrogate in 30/32. The direct Random-minus-Surrogate correlation ranges from -0.502 to -0.057, is negative in 32/32 and significantly negative in 28/32 after Holm correction. These controls support a policy-agnostic, retrospective interpretation of headroom.

10 Vietnamese Best-of-K Curves

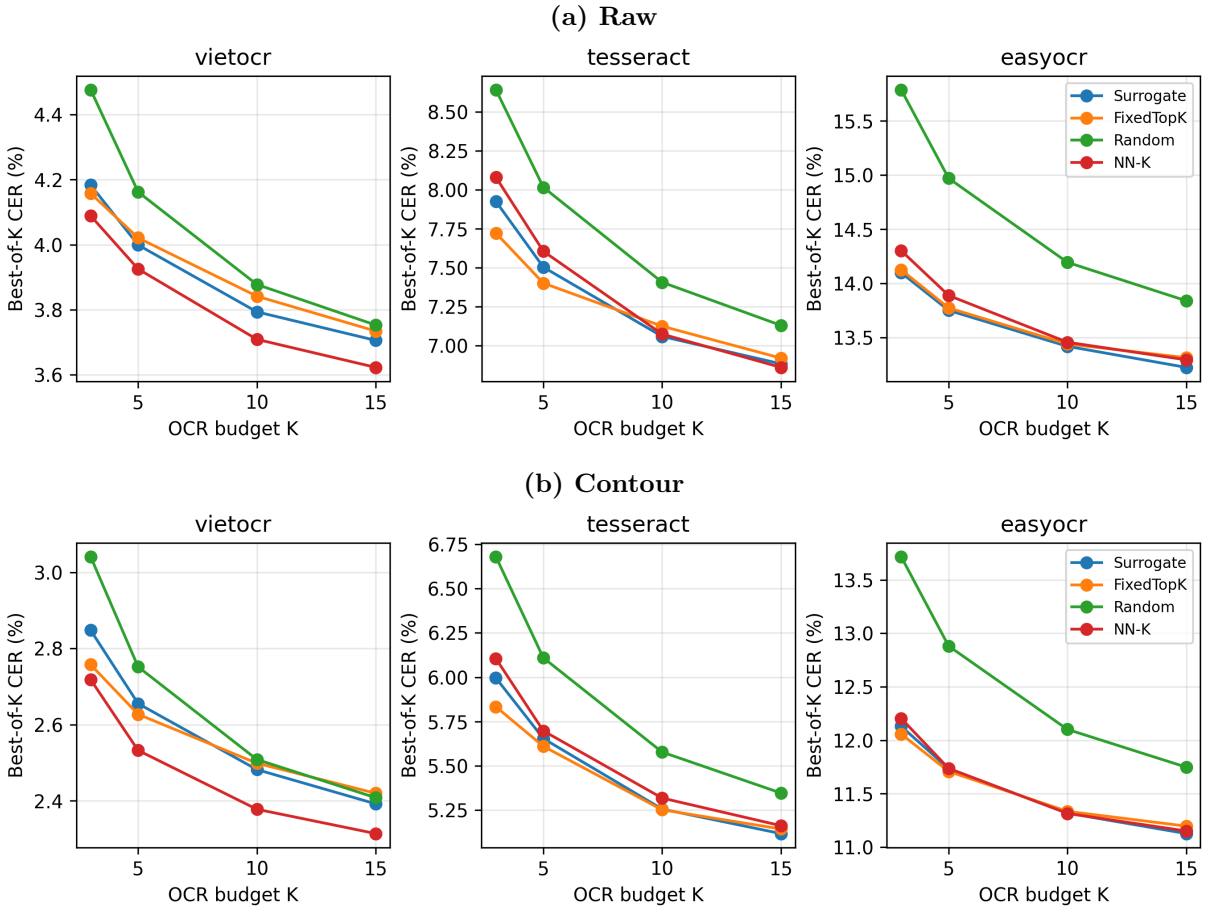


Fig. 1 Mean best-of-K CER for Surrogate, FixedTopK, Random, and NN-K on Vietnamese administrative documents: (a) raw pages and (b) contour-suppressed pages. Surrogate/Random values match main-manuscript Table 1, and FixedTopK values at $K = 3, 15$ match Table 2; all plotted values are available in the public tables. Each recognizer panel uses its own y-axis scale

11 Multiplicity-Adjusted Best-of-K Policy Comparisons

For best-of-K inference, each policy contrast within one evaluation block is a Holm family: 12 recognizer-budget tests in each Vietnamese block and eight tests on SROIE. Table 8 reports direction-specific counts before and after correction. "A lower" means lower mean CER for the first policy named in the contrast; "B lower" means lower mean CER for the second.

Table 8: Direction-specific significant comparison counts. Each cell is unadjusted/Holm-adjusted; Holm correction is within the block and contrast shown.

Block	Contrast (A vs B)	A lower	B lower
Raw VN	Surrogate vs Random	12/12	0/0
Raw VN	Surrogate vs FixedTopK	1/0	0/0
Raw VN	NN-K vs Random	12/12	0/0
Raw VN	NN-K vs FixedTopK	2/1	4/0
Raw VN	NN-K vs Surrogate	3/0	3/0
Contour VN	Surrogate vs Random	12/12	0/0
Contour VN	Surrogate vs FixedTopK	1/1	2/1
Contour VN	NN-K vs Random	12/12	0/0
Contour VN	NN-K vs FixedTopK	2/2	2/0
Contour VN	NN-K vs Surrogate	3/3	2/0
SROIE	Surrogate vs Random	8/8	0/0
SROIE	Surrogate vs FixedTopK	8/8	0/0
SROIE	NN-K vs Random	8/8	0/0
SROIE	NN-K vs FixedTopK	8/8	0/0
SROIE	NN-K vs Surrogate	2/2	4/3

At $K = 15$, EasyOCR has 215/300 zero Surrogate–FixedTopK differences in raw and 216/300 in contour. Table 9 reports a post hoc zero-handling check using separate 12-test families for each block and convention. The specified `wilcox` analysis remains primary; its apparent advantage in contour, and the advantages found under other conventions in raw, are sensitive to zero handling. The adverse contour VietOCR $K = 3$ result remains significant under all three checked conventions.

Table 9: EasyOCR $K = 15$ zero-handling sensitivity. Differences are Surrogate minus FixedTopK (pp). Holm correction uses 12 tests per block and convention.

Block	Zero method	Zero pages	Diff	Holm p
Raw	wilcox	215	-0.09	0.0626
Raw	pratt	215	-0.09	0.00277
Raw	zsplit	215	-0.09	0.0257
Contour	wilcox	216	-0.07	0.0384
Contour	pratt	216	-0.07	0.0699
Contour	zsplit	216	-0.07	0.234

12 SROIE Letter-Case Metric Transition and Case-Sensitive Sensitivity

The primary SROIE results in this revision use casefolded CER. The original analysis was case-sensitive, and the metric decision was changed after initial results and the casefold best-of-K sensitivity were available. Inspection of letter-case fidelity motivated the change; it was not prespecified before those results. Rotman et al. [2] likewise use case-insensitive edit-distance evaluation for SROIE because its annotations do not preserve letter case, although their evaluation is not identical to ours. Baek et al.

[1] report case-related annotation inconsistencies in other scene-text benchmarks; that observation does not establish annotation errors in SROIE.

Both predictions and references receive the same NFC and whitespace normalization, with Unicode casefold additionally applied for SROIE. Accents and punctuation are retained. Vietnamese evaluation remains case-sensitive on corrected v3 references. Surrogate targets, train-only static rankings, NN-K transfer choices and all nested SROIE selector/control training targets were updated consistently. Selector text-composition and agreement features are computed from the original OCR strings, not casefolded strings.

Tables 10 and 11 compare both metrics on the same corrected 612 receipts, 50 candidates and new outer folds. Earlier 616-page results remain historical, not the numerical source of these tables. Surrogate and NN-K outperform Random and FixedTopK in 8/8 comparisons under both metrics. Surrogate is significantly better than NN-K for Tesseract $K = 3, 5, 10$ under both metrics. NN-K's EasyOCR $K = 5$ advantage is significant under both metrics; its $K = 3$ advantage is significant under casefolding ($p_H = 0.0336$) but not case-sensitive CER ($p_H = 0.0778$). Crossing a significance threshold is not a formal metric-by-policy interaction test, nor does nonsignificance establish equivalence.

Each metric/contrast has a separate eight-test Holm family using two-sided Wilcoxon and the `wilcox` zero convention. Mean-difference CIs are per-comparison 95% paired percentile intervals, with 10,000 resamples and recorded seeds, not multiplicity-adjusted. Nonzero rounded-zero endpoints retain their signs. Significance statements use Wilcoxon–Holm, not CI zero exclusion; for example, the Tesseract $K = 10$ NN-K–Surrogate mean-difference intervals contain zero under both metrics despite significant signed-rank tests. Historical BCa and multiple-seed diagnostics on 616 receipts remain archived but were not repeated for these 612-page intervals and do not validate their coverage.

Table 10: Matched 612-receipt best-of-K sensitivity: Holm-significant lower-CER counts. Each metric/contrast has a separate eight-test family; A/B identifies the lower-CER policy.

Contrast (A vs B)	Case-sensitive A/B	Casefold A/B
Surrogate vs Random	8/0	8/0
Surrogate vs FixedTopK	8/0	8/0
NN-K vs Random	8/0	8/0
NN-K vs FixedTopK	8/0	8/0
NN-K vs Surrogate	1/3	2/3

Table 11: NN-K minus Surrogate on the same 612 receipts and folds (pp). CS: case-sensitive sensitivity; CF: primary casefold. CIs are per-comparison 95% paired percentile intervals, not multiplicity-adjusted; p_H uses each metric's eight-test family. Nonzero rounded-zero endpoints retain their signs. Significance follows Wilcoxon–Holm, not mean-CI zero exclusion.

Engine	K	CS Δ	CS 95% CI	CS p_H	CF Δ	CF 95% CI	CF p_H
Tesseract	3	+0.53	[+0.21, +0.86]	0.00164	+0.61	[+0.28, +0.93]	0.000261
Tesseract	5	+0.34	[+0.09, +0.59]	0.00164	+0.38	[+0.14, +0.62]	0.00254
Tesseract	10	+0.16	[-0.04, +0.35]	0.00646	+0.19	[-0.01, +0.40]	0.0343
Tesseract	15	+0.03	[-0.12, +0.18]	0.151	+0.09	[-0.07, +0.26]	0.125
EasyOCR	3	-0.16	[-0.27, -0.05]	0.0778	-0.16	[-0.28, -0.03]	0.0336
EasyOCR	5	-0.13	[-0.22, -0.05]	0.0372	-0.13	[-0.23, -0.04]	0.0191
EasyOCR	10	-0.02	[-0.08, +0.04]	1.00	-0.03	[-0.11, +0.04]	0.866
EasyOCR	15	+0.03	[-0.03, +0.08]	1.00	-0.01	[-0.07, +0.05]	0.798

This cross-metric sensitivity directly compares best-of-K outputs only. Current casefold one-output selector results are reported below; historical case-sensitive selector artifacts are retained separately, but no formal paired cross-metric selector sensitivity is claimed.

13 Policy-Matched Oracle-Free Selector Comparisons

We compared complete one-output policies using strictly nested Surrogate, NN-K and FixedTopK top-3 selectors. The 16 full-feature comparisons form one Holm family. Surrogate is significantly better than NN-K on raw Tesseract and SROIE Tesseract, and better than FixedTopK on SROIE Tesseract. It is significantly worse than FixedTopK on contour VietOCR. The other 12 rows are not significant. Differences are Surrogate minus comparator in pp.

The 95% bootstrap intervals in this and the rank-only table are pointwise intervals for the paired mean CER difference, not Holm-adjusted simultaneous intervals. Statistical-significance claims use the Holm-adjusted Wilcoxon p-values; the signed-rank test does not directly test a null hypothesis about the mean difference.

Table 12: Strict policy-matched selector comparison. Holm p-values are adjusted across all 16 rows. CIs are per-comparison 95% intervals, not multiplicity-adjusted. Nonzero bounds retain their signs when rounded: +0.00 or −0.00 does not mean exactly zero. Significance follows Holm-adjusted Wilcoxon tests, not CI zero exclusion.

Setting	Recognizer	Comparator	Surrogate	Comparator CER	Diff	95% CI	Holm p
Raw	VietOCR	NN-K	4.48	4.55	-0.07	[-0.20, +0.07]	1.00
Raw	Tesseract	NN-K	8.57	9.07	-0.50	[-0.77, -0.23]	0.0256
Raw	EasyOCR	NN-K	14.99	15.27	-0.28	[-0.51, -0.06]	0.236
Contour	VietOCR	NN-K	3.22	3.17	+0.05	[-0.08, +0.18]	1.00
Contour	Tesseract	NN-K	6.68	6.87	-0.19	[-0.42, +0.04]	0.821
Contour	EasyOCR	NN-K	12.97	13.20	-0.23	[-0.46, -0.01]	0.438
SROIE	Tesseract	NN-K	25.15	25.59	-0.45	[-0.97, +0.19]	0.0138
SROIE	EasyOCR	NN-K	19.78	19.67	+0.11	[-0.12, +0.32]	0.304
Raw	VietOCR	FixedTopK	4.48	4.40	+0.08	[-0.04, +0.21]	0.611
Raw	Tesseract	FixedTopK	8.57	8.42	+0.16	[-0.06, +0.38]	1.00
Raw	EasyOCR	FixedTopK	14.99	14.93	+0.06	[-0.13, +0.25]	1.00
Contour	VietOCR	FixedTopK	3.22	3.00	+0.22	[+0.10, +0.34]	0.0441
Contour	Tesseract	FixedTopK	6.68	6.50	+0.18	[-0.05, +0.42]	1.00
Contour	EasyOCR	FixedTopK	12.97	12.85	+0.12	[-0.06, +0.29]	0.821
SROIE	Tesseract	FixedTopK	25.15	26.33	-1.18	[-1.55, -0.83]	2.73×10^{-9}
SROIE	EasyOCR	FixedTopK	19.78	19.91	-0.13	[-0.29, +0.03]	1.00

14 Rank-Only Selector Sensitivity Analysis

The main Surrogate selector uses continuous predictions and margins, whereas NN-K and FixedTopK provide an order. Replacing these prediction-related features by ranks preserves the exact surrogate top-3 candidates and OCR outputs before refitting the nested selector. CER changes by −0.059 to +0.035 pp relative to full features. The separate 16-test sensitivity family has three significant advantages (NN-K on raw/SROIE Tesseract, FixedTopK on SROIE Tesseract) and no significant disadvantages. This does not isolate shortlist quality because the selector is refit and retains OCR-output features.

Table 13: Rank-only Surrogate selector sensitivity. Holm p-values are adjusted across all 16 rows in this separate sensitivity family. CIs are per-comparison 95% intervals, not multiplicity-adjusted; signs of nonzero rounded endpoints are retained. Significance follows Holm-adjusted Wilcoxon tests.

Setting	Recognizer	Comparator	Rank-only Surr.	Comparator CER	Diff	95% CI	Holm p
Raw	VietOCR	NN-K	4.47	4.55	-0.08	[-0.21, +0.05]	1.00
Raw	Tesseract	NN-K	8.58	9.07	-0.49	[-0.77, -0.22]	0.0104
Raw	EasyOCR	NN-K	15.02	15.27	-0.25	[-0.47, -0.02]	0.380
Contour	VietOCR	NN-K	3.23	3.17	+0.06	[-0.07, +0.20]	1.00
Contour	Tesseract	NN-K	6.68	6.87	-0.19	[-0.42, +0.03]	0.939
Contour	EasyOCR	NN-K	12.91	13.20	-0.29	[-0.52, -0.05]	0.115
SROIE	Tesseract	NN-K	25.13	25.59	-0.46	[-0.98, +0.18]	0.00797
SROIE	EasyOCR	NN-K	19.74	19.67	+0.07	[-0.15, +0.27]	1.00
Raw	VietOCR	FixedTopK	4.47	4.40	+0.07	[-0.05, +0.18]	1.00
Raw	Tesseract	FixedTopK	8.58	8.42	+0.16	[-0.07, +0.39]	1.00
Raw	EasyOCR	FixedTopK	15.02	14.93	+0.09	[-0.09, +0.28]	1.00
Contour	VietOCR	FixedTopK	3.23	3.00	+0.23	[+0.10, +0.36]	0.105
Contour	Tesseract	FixedTopK	6.68	6.50	+0.17	[-0.05, +0.40]	1.00
Contour	EasyOCR	FixedTopK	12.91	12.85	+0.06	[-0.13, +0.24]	1.00
SROIE	Tesseract	FixedTopK	25.13	26.33	-1.20	[-1.56, -0.85]	1.25×10^{-9}
SROIE	EasyOCR	FixedTopK	19.74	19.91	-0.18	[-0.34, -0.00]	0.187

15 SROIE Intermediate-Budget Results

Table 14: SROIE receipt-setting intermediate-budget CER results. Values are percentages. These rows complement the main-manuscript SROIE table, which reports $K = 3$ and $K = 15$.

Recognizer	K	Surrogate K	NN-K	FixedTopK	Random K
Tesseract	5	23.11	23.49	24.33	26.18
Tesseract	10	22.16	22.35	22.95	24.01
EasyOCR	5	18.52	18.39	18.85	19.82
EasyOCR	10	18.17	18.14	18.38	18.90

16 Post Hoc Standalone-Feature Sensitivity

The legacy `orig_noise` column is the standard deviation of the Laplacian response, the square root of `orig_blur`, rather than an independent noise estimate. After the primary results were available, a supplementary best-of-K protocol was fixed before its fits: rerun the original input (Full: 22 surrogate and 12 NN-K columns) and remove only this standalone column (Drop: 21 and 11 columns). The interaction `inter_noise_bilateral_d` was retained unchanged and means $\sqrt{\text{Laplacian variance}} \times$ `bilateral_d`. Both conditions used the same frozen references, CER rules, five page-group folds, candidate order, model definitions and seed 42. Nested surrogate selection and NN-K were rerun; strict one-output selectors were not. The primary 22/12-column results were retained regardless of this supplementary outcome. Removal can change model-family selection, stochastic column subsampling and NN distance weighting; this is not an isolated measurement of information content.

The original Vietnamese Full reruns reproduced primary per-page CER and model-family selections. For the corrected SROIE cohort, A and primary analyses reuse the identical verified Full outputs and new folds; Drop was fitted on that same cohort. Full reproduces all 160 primary policy-comparison rows. Direct Drop-minus-Full comparisons for Surrogate and NN-K used six separate policy-by-block Holm families (12 tests per Vietnamese block and eight for SROIE). The five existing policy contrasts were also recomputed under each condition using their original 15 block-by-contrast families. Tests were paired, two-sided Wilcoxon with `zero_method=wilcox`. Conditional paired-page percentile 95% CIs used 10,000 resamples with fixed recorded seeds; they are not multiplicity-adjusted and do not include refit, seed or split variability.

Table 15 summarizes mean CER changes. None of the 64 direct Drop-minus-Full tests survives Holm correction. This does not establish equivalence or absence of a feature-removal effect; full-precision differences and unadjusted mean-difference CIs are released.

Seven of the 160 policy significance classifications changed (Table 16), without reversing the mean-difference signs of those seven rows. Six Vietnamese comparisons lost significance, including both the contour EasyOCR $K = 15$ Surrogate advantage and the contour VietOCR $K = 3$ Surrogate disadvantage relative to FixedTopK. One SROIE comparison gained significance: Surrogate over NN-K for Tesseract $K = 15$. Five other, nonsignificant Vietnamese mean contrasts reversed sign; all ten associated adjusted p-values equal 1. In both conditions, Surrogate and NN-K remained better than Random after Holm in all 24 Vietnamese and eight SROIE comparisons, and both remained better than FixedTopK in all eight SROIE comparisons. Secondary policy distinctions are nevertheless sensitive to the feature configuration; neither non-significance nor an unchanged classification establishes equivalence. A change in significance is not a formal test of the difference between contrasts. These best-of-K results do not establish strict-selector robustness. Full-precision paired values, all 384 comparisons, the original design protocol, corrected-cohort provenance and arithmetic replay are included in the public supplement.

17 Contour Suppression: Schematic and Threshold Check

The exploratory threshold check used VietOCR on the first three lexicographically sorted available PNG filenames from each of the ten Vietnamese year folders (30 pages total). Its saved report is dated 23 August 2026, before the later reference-NFC correction. It was not rerun on the final references or with Tesseract and EasyOCR.

Table 15: Standalone Laplacian-standard-deviation removal: range of Drop-minus-Full mean best-of-K CER differences across engines and budgets (pp). Negative is lower CER after removal. The final column counts Holm-significant lower/higher results in each separate direct-comparison family. All 64 comparisons, including CIs, are released.

Setting	Policy	Minimum	Maximum	Tests	Lower/Higher
VN contour	NN-K	-0.03	+0.02	12	0/0
VN contour	Surrogate	-0.04	+0.04	12	0/0
VN raw	NN-K	-0.05	+0.02	12	0/0
VN raw	Surrogate	-0.06	+0.04	12	0/0
SROIE	NN-K	-0.06	+0.07	8	0/0
SROIE	Surrogate	-0.05	+0.34	8	0/0

Table 16: All 7 policy comparisons whose Holm-adjusted significance classification changes after removing only the standalone column. Differences are first policy minus second (pp); negative favors the first. p_H uses the same prespecified block/contrast family in each condition (12 tests for each Vietnamese block, eight for SROIE). These threshold crossings are not formal tests of a difference between contrasts. Full-precision results for all 160 comparisons per condition are released.

Setting	Engine	K	Contrast	Full Δ	Drop Δ	Full p_H	Drop p_H
VN contour	EasyOCR	15	Surr.–FixedTopK	-0.07	-0.07	0.0384	0.104
VN contour	VietOCR	3	Surr.–FixedTopK	+0.09	+0.09	0.0361	0.181
VN contour	VietOCR	10	NN-K–FixedTopK	-0.12	-0.12	0.0445	0.0507
VN contour	VietOCR	10	NN-K–Surr.	-0.10	-0.09	0.0371	0.138
VN contour	VietOCR	15	NN-K–FixedTopK	-0.11	-0.10	0.0189	0.0507
VN raw	VietOCR	10	NN-K–FixedTopK	-0.13	-0.13	0.0355	0.0656
SROIE	Tesseract	15	NN-K–Surr.	+0.09	+0.15	0.125	0.0306

Table 17 summarizes one-at-a-time threshold changes. The lower aspect-ratio bound remained fixed at 1.0 and was not tested. These are descriptive changes in sample mean CER, not per-page maxima or significance tests; they do not establish robustness on the final references.

Table 17: Historical contour-threshold check on 30 pages with VietOCR. Default mean CER was 2.72%. Each threshold was varied separately, with the others fixed at their defaults (area 500 square pixels, upper aspect ratio 5, width 80 pixels, height 30 pixels). Changes are relative to that default and expressed in percentage points.

Threshold	Tested values	Max $\lvert\Delta$mean CER$\rvert$ (pp)
Area (px^2)	500, 1000, 1500, 2000, 3000	0.56
Upper aspect ratio	3, 4, 5, 6, 8	0.06
Minimum width (pixels)	50, 60, 80, 100, 120	0.25
Minimum height (pixels)	15, 20, 30, 40, 50	0.29

References

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[2] Daniel Rotman, Ophir Azulai, Inbar Shapira, Yevgeny Burshtein, and Udi Barzelay. Detection masking for improved ocr on noisy documents, 2022. arXiv:2205.08257.

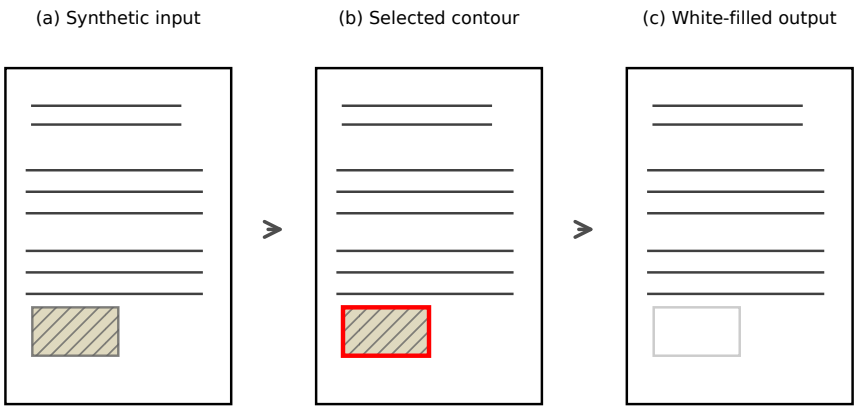


Fig. 2 Synthetic illustration of contour suppression. A stamp-like rectangular artifact is identified and white-filled before OCR. The figure is schematic and does not reproduce a real source document